EXP - 07: PRODUCER CONSUMER PROBLEM SOLUTION USING SEMAPHORE

```c
#include<stdio.h>
#include<pthread.h>
#include<semaphore.h>
#define BUFFER_SIZE 5
int buffer[BUFFER_SIZE];
int in = 0;
int out = 0;
sem_t empty;
sem_t full;
sem_t mutex;
void *producer(void *arg){
        int item;
        for(int i=0;i<BUFFER_SIZE;i++){
                item =i;
                sem_wait(&empty);
                sem_wait(&mutex);
                buffer[in] = item;
                printf("Producer: %d\n",item);
                in =(in +1)%BUFFER_SIZE;
                sem_post(&mutex);
                sem_post(&full);
        }
}
void *consumer(void*arg){
        int item;
        for(int i=0;i<BUFFER_SIZE;i++){
                sem_wait(&full);
                sem_wait(&mutex);
                item = buffer[out];
                printf("Consumed:%d\n",item);
                out = (out + 1) % BUFFER_SIZE;
                sem_post(&mutex);
                sem_post(&empty);
        }
}
int main(){
        pthread_t p,c;
        sem_init(&empty, 0 , BUFFER_SIZE);
        sem_init(&full,0, 0);
        sem_init(&mutex,0, 1);
        pthread_create(&p, NULL, producer,NULL);
        pthread_create(&c,NULL,consumer,NULL);
        pthread_join(p,NULL);
        pthread_join(c,NULL);
        sem_destroy(&empty);
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```

```c
int in = 0;
int out = 0;
sem_t empty;
sem_t full;
sem_t mutex;
void *producer(void *arg){
        int item;
        for(int i=0;i<BUFFER_SIZE;i++){
                item =i;
                sem_wait(&empty);
                sem_wait(&mutex);
                buffer[in] = item;
                printf("Producer: %d\n",item);
                in =(in +1)%BUFFER_SIZE;
                sem_post(&mutex);
                sem_post(&full);
        }
}
void *consumer(void*arg){
        int item;
        for(int i=0;i<BUFFER_SIZE;i++){
                sem_wait(&full);
                sem_wait(&mutex);
                item = buffer[out];
                printf("Consumed:%d\n",item);
                out = (out + 1) % BUFFER_SIZE;
                sem_post(&mutex);
                sem_post(&empty);
        }
}
int main(){
        pthread_t p,c;
        sem_init(&empty, 0 , BUFFER_SIZE);
        sem_init(&full,0, 0);
        sem_init(&mutex,0, 1);
        pthread_create(&p, NULL, producer,NULL);
        pthread_create(&c,NULL,consumer,NULL);
        pthread_join(p,NULL);
        pthread_join(c,NULL);
        sem_destroy(&empty);
        sem_destroy(&full);
        sem_destroy(&mutex);
        return 0;
}
                                                                    50,0-1        Bot
```

```
[rene@localhost producerconsumer]$ vi pc.c
[rene@localhost producerconsumer]$ gcc pc.c
[rene@localhost producerconsumer]$ ./a.out
Producer: 0
Producer: 1
Producer: 2
Producer: 3
Producer: 4
Consumed:0
Consumed:1
Consumed:2
Consumed:3
Consumed:4
[rene@localhost producerconsumer]$
```